Joanna Zou, PhD

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Education

Massachusetts Institute of Technology — Cambridge, MA
PhD in Computational Science & Engineering — Feb. 2022 – May 2026
- Advisor: Youssef Marzouk
- Thesis: *Goal-Oriented Learning of Stochastic Dynamical Systems*

Stanford University — Stanford, CA
MS in Structural Engineering — Sep. 2018 – May 2020
- Advisor: Gregory Deierlein
- Funding: Stanford Blume Center Graduate Fellowship

Columbia University — New York, NY
BS in Civil Engineering — Aug. 2014 – May 2018
- GPA: 3.95 / 4.00 (*magna cum laude*)
- Top graduate in civil engineering class
- Funding: Henry Michel Scholarship

Employment

Massachusetts Institute of Technology — Cambridge, MA
Postdoctoral Associate — Jun. 2026 – Present

Methodological work:
- Subspace optimization solutions to stochastic optimal control and reverse diffusion sampling problems with improved convergence rates
- Reduced-dimensional parameter calibration of stochastic systems using path-space sensitivity

Graduate Research Assistant — Feb. 2022 – May 2026

Methodological work:
- Interacting particle systems for active learning of energy-based models in molecular dynamics, validated on MACE force fields of silicon and alanine dipeptide [C5]
- Goal-oriented loss for learning surrogate SDEs with error bounds on path-space statistics, validated on learning neural network drifts [PP7]
- Reduced-order representations of SDEs with uncertain drift functions for uncertainty quantification of path-space statistics, validated using finite element discretizations [PP8]
- Determinantal point processes for probabilistic D-optimal experimental design, evaluated using invariant kernels of descriptors of molecular configurations [C4]
- Sequential importance sampling in controlled generative diffusion models for rare-event simulation

Experience:
- Develop custom Julia packages for Bayesian inference and active learning of ML force fields
- Construct HPC workflows for neural network training in PyTorch and Monte Carlo simulations
- Coordinate weekly journal club for reviewing papers on topics in applied math, statistics, and ML
- Tutor students in undergraduate courses on probability and statistics

University of Potsdam Potsdam, Germany
Visiting PhD Research Fellow in Mathematics Jun. 2024 – Aug. 2024
Methodological work:

- Variational objectives for improved sample efficiency and fidelity of transition properties of SDEs, with derived analytical gradients for optimization

Sandia National Laboratories Livermore, CA
Graduate Research Intern Jun. 2023 – Aug. 2023
Methodological work:

- Multifidelity importance sampling-based control variate framework to reduce the cost of uncertainty quantification of ML force fields, validated using LAMMPS molecular simulation

Technical University of Delft Delft, Netherlands
Fulbright Research Fellow Jan. 2021 – Dec. 2021
Methodological work:

- Latent variable model to predict strain-induced damage in offshore wind turbines using sparse acceleration data and Bayesian filtering, validated on industry datasets

NHERI-SimCenter Berkeley, CA
Software Development & Research Associate Aug. 2020 – Dec. 2020
Experience:

- Develop Python software for natural hazards simulation utilizing modern ML and UQ workflows
- Construct parallel computing workflow for multiscale earthquake simulation

Stanford University Stanford, CA
Graduate Research Assistant Sep. 2018 – May 2020
Methodological work:

- Heteroscedastic Gaussian process surrogate models of the seismic response of buildings using recorded ground motions, developed to accelerate probabilistic risk analysis [C2]
- SVM-based gait anomaly classifier using signal processing-based features of footstep vibrations [C1]
- Comparison of U-Net CNNs and LSTMs for land use classification from Sentinel-1 satellite imagery

Arup San Francisco, CA
Advanced Technology & Research Intern Jun. 2018 – Aug. 2018
Experience:

- Perform dynamic finite element analysis for soil-structure interaction modeling in LS-DYNA
- Conduct probabilistic risk analysis of building portfolio in accordance with FEMA P-58 guidelines

Arup Edison, NJ
Structural Engineering Intern Jun. 2017 – Aug. 2017
Experience:

- Design concrete decks for airport terminal platforms and hurricane-retrofitted structures

Carleton Strength of Materials Laboratory New York, NY
Lab Assistant Aug. 2015 – Dec. 2015
Experience:

- Conduct tensile strength tests on steel reinforcement bars sampled from operational bridges

Publications

Preprints

[PP8] **J. Zou**, H. C. Lie, Y. Marzouk. "Derivative-Informed Dimension Reduction of Stochastic Differential Equations: A Path-Space Sensitivity Approach." *Preprint, under review at SIAM/ASA Journal of Uncertainty Quantification. arXiv:2610.33210*. 2026.

[PP7] **J. Zou**, H. C. Lie, Y. Marzouk. "Goal-oriented learning of stochastic differential equations using error bounds on path-space observables." *Preprint, under review at Multiscale Modeling and Simulation (SIAM). arXiv:2603.20467*. 2026.

Journal articles

[J6] **J. Zou**, E. Lourens, A. Cicirello. "Virtual sensing of subsoil strain response in monopile-based offshore wind turbines via Gaussian process latent force models." *Mechanical Systems & Signal Processing*. 200, 110488. 2023.

Peer-reviewed conference proceedings

[C5] **J. Zou**, F. Birks, D. Foster, Y. Marzouk. "Stein kernelized molecular dynamics for active learning of interatomic potentials." *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.

[C4] **J. Zou**, Y. Marzouk. "Data curation for machine learning interatomic potentials by determinantal point processes." *ICLR AI for Accelerated Materials Design Workshop*. 2025.

[C3] **J. Zou**, A. Cicirello, A. Iliopoulos, E. Lourens. "Gaussian process latent force models for virtual sensing in a monopile-based offshore wind turbine." *Proceedings of the European Workshop on Structural Health Monitoring, pp. 290-298.* 2022.

[C2] **J. Zou**, D. Welch, A. Zsarnoczay, A. Taflanidis, G. Deierlein. "Surrogate modeling for the seismic response estimation of residential wood frame structures." *Proceedings of the 17th World Conference in Earthquake Engineering*. 2020.

[C1] Y. Dong, **J. Zou**, J. Liu, J. Fagert, M. Mirshekari, L. Lowes, M. Iammarino, P. Zhang, H. Noh. "Physics-informed analysis of patient induced structural vibration data for early muscular dystrophy detection." *Third Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing (Ubicomp-CPD), pp. 525-531*. 2020.

Talks and presentations

Contributed talks

[T16] "Stein kernelized molecular dynamics for fine-tuning machine learning interatomic potentials." *International Conference on Multiscale Materials Modeling*, Jeju, Korea. November 2026.

[T15] "Derivative-informed dimension reduction for the uncertainty quantification of path-space statistics of SDEs." *Workshop on Mathematical and Computational Foundations of Digital Twins*, Corsica, France. November 2026.

[T14] "Dimension reduction on Hilbert spaces of drift functions of stochastic differential equations." *SIAM Conference on Uncertainty Quantification (SIAM UQ)*, Minneapolis, MN. March 2026.

[T13] "Goal-oriented learning of stochastic dynamical systems using error bounds on path-space observables." *SIAM Conference on Applied Dynamical Systems (SIAM DS)*, Denver, CO. May 2025.

[T12] “Path-space error bounds for the statistical learning of stochastic differential equations.” *University of Potsdam SIAM Chapter Seminar*, Potsdam, Germany. August 2024.

[T11] “Cairn.jl: Enhanced Molecular Dynamics for Active Learning.” *JuliaCon*, Eindhoven, Netherlands. July 2024.

[T10] “Active learning of energy-based models via stochastic Stein variational gradient descent.” *SIAM Conference on Uncertainty Quantification (SIAM UQ)*, Trieste, Italy. February 2024.

[T9] “Adaptive importance sampling for gradient-based dimension reduction in stochastic systems.” *International Conference on Uncertainty Quantification in Computational Science and Engineering (UNCECOMP)*, Athens, Greece. June 2023.

[T8] “Gaussian process latent force models for virtual sensing of offshore wind turbines.” *European Workshop on Structural Health Monitoring (EWSHM)*, Sicily, Italy. June 2022.

Poster presentations

[P7] “Stein kernelized molecular dynamics for machine learning of interatomic potentials.” *Advances in Neural Information Processing Systems (NeurIPS)*, Atlanta, Georgia. December 2026.

[P6] “Goal-oriented learning and uncertainty quantification of stochastic differential equations.” *MIT Center for the Exascale Simulation of Coupled High Enthalpy Fluid-Structure Interactions (CHEFSI)*, Cambridge, MA. April 2026.

[P5] “A goal-oriented loss for learning transition times with machine learning potentials.” *ICMS Workshop on Computational Materials Science and Mathematics at the Particle and Atomistic Scales*, Edinburgh, UK. November 2025.

[P4] “Data curation for machine learning interatomic potentials by determinantal point processes.” *ICLR AI for Accelerated Materials Design Workshop*, Singapore. April 2025.

[P3] “Using error bounds on molecular properties for goal-oriented learning of machine learning potentials.” *MIT Center for the Exascale Simulation of Material Interfaces in Extreme Environments (CESMIX)*, Cambridge, MA. April 2025.

[P2] “Active subspaces for Bayesian inference of machine learning interatomic potential models.” *MIT Center for the Exascale Simulation of Material Interfaces in Extreme Environments (CESMIX)*, Cambridge, MA. Oct. 2023.

[P1] “Bayesian modeling and active learning for molecular dynamics: fitting machine learning interatomic potentials to data.” *EPFL CECAM Workshop on Error Control in First Principles Modeling*, Lausanne, Switzerland. June 2022.

Service

Minisymposium organization

SIAM Conference on Uncertainty Quantification (SIAM UQ)

“Advances in MCMC Sampling Methods” 2026

“UQ for Multiscale Modeling in Computational Chemistry” 2026

SIAM Conference on Applied Dynamical Systems (SIAM DS)
"Learning with Dynamical Systems and Their Observable Statistics" 2025

Peer reviews
ICLR Workshop on Physics-Inspired AI for Scientific Discovery
International Journal of Uncertainty Quantification (IJUQ)
Mechanical Systems and Signal Processing (MSSP)
Data-Centric Engineering (DCE) Journal

Student organizations

MIT Graduate Student Council, External Affairs Committee
Science Policy Advocate 2025
- Propose policy items on federal research funding and graduate education to U.S. congressional offices in Washington, D.C.

MIT Assoc. of Computational Engineering and Science Students (ACSES)
Vice President 2023-2025
- Increase engagement and seminar participation among CSE graduate students

MIT European Club
Outreach Chair 2024-2025
- Lead outreach efforts to recruit technology start-ups, companies, universities, and research institutions to participate in the MIT European Career Fair

Columbia American Society of Civil Engineers (ASCE)
President 2017-2018
- Institute undergraduate mentorship program, panel talks, and career development workshops in collaboration with professional engineering societies

Teaching & mentoring

MIT AeroAstro Graduate Admissions Assistance Program
Mentor 2022-2025
- Provide 1-on-1 guidance on PhD applications for prospective students

Stanford Academic Resources Center
Course Tutor and Student Mentor 2020
- Provide 1-on-1 tutoring for Bachelor/Master courses in math, statistics, physics, and computer science

Stanford CEE 280: Advanced Structural Analysis
Guest Lecturer 2019
- Design and deliver a 2-part lecture and homework assignment on numerical methods for convex optimization to a 40-student graduate course

Columbia University Office of Residential Programs
Resident Advisor 2015-2018
- Advised, mediated, and supervised 50-70 students per year in undergraduate residences

Honors and scholarships

SFB 1294 Data Assimilation PhD Research Fellowship	2024
JuliaCon Young Researchers Grant	2024
CECAM Workshop Best Poster Award	2022
Fulbright U.S. Student Scholarship *(nationally awarded 1-year research fellowship)*	2020
Stanford Blume Center Graduate Fellowship *(full funding for Masters degree)*	2018
Thornton-Tomasetti Foundation Scholarship *(nationally awarded scholarship for graduate studies)*	2018
ASCE Arthur Tuttle Fellowship	2018
Columbia George Wendell Memorial Award *(academic excellence and service award)*	2018
Columbia Robert Ridgway Award *(top student in civil engineering class)*	2018
ACEC New York Merit Scholarship	2017
Columbia University Henry Michel Scholarship	2015
National Merit Scholarship	2014